■ DATA SCIENCE / AI | PROJECT REPORT

SMART ELECTRICITY CONSUMPTION ANALYSIS AND PREDICTION

A PROJECT REPORT SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE COURSE

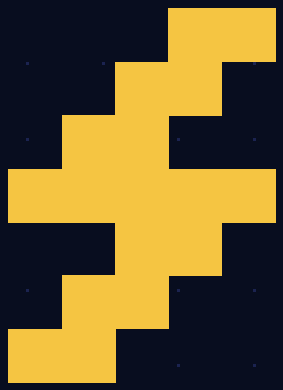
DATA SCIENCE AND ARTIFICIAL INTELLIGENCE

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August-December 2026

CERTIFICATE

This is to certify that the project entitled **"SMART ELECTRICITY CONSUMPTION ANALYSIS AND PREDICTION"** is a bona fide work carried out by **Ishan Ahirwar (261AI027)**, **Arjun S M (261AI012)**, **Neelesh Reddy (261AI036)**, **Shuban Kamath (261AI058)**, **Mourya G N (261AI040)**, **Srujan Soraganvi (261AI060)** under my supervision and guidance in partial fulfilment of the requirements for the course **Data Science and Artificial Intelligence**, Department of Information Technology, National Institute of Technology Karnataka, August-December 2026.

PROJECT GUIDE	**HEAD OF DEPARTMENT**

DATE: _____ PLACE: _____

DECLARATION

We hereby declare that the project entitled **"SMART ELECTRICITY CONSUMPTION ANALYSIS AND PREDICTION"** is our original work carried out under the guidance of **Dr. Suresan Pareth**. This work has not been submitted elsewhere for the award of any degree or diploma, except where acknowledged in the report.

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We also thank everyone who contributed to the successful completion of this work.

ABSTRACT

Electricity consumption varies with time, weather, household requirements and industrial activity. Analysing historical usage helps us understand consumption patterns and estimate future demand. This project analyses 5,000 half-hourly smart-meter readings (1 January to 14 April 2024) using statistical and machine learning techniques.

The data are checked for missing values, duplicates and timing gaps, summarised with descriptive statistics, and examined for correlations. Regression, time-series and anomaly-detection methods are compared against simple baselines, and results are visualised as daily, weekly and monthly patterns.

Only the average past consumption is meaningfully related to consumption ($r = 0.32$); weather variables are essentially uncorrelated. Linear Regression is the best regression model ($R^2 = 0.24$, RMSE 13% below the mean baseline), and a supervised classifier detects labelled anomalies with an F1-score of 0.92. The work supports energy monitoring, consumption planning and efficient energy management.

Keywords: Electricity Consumption, Energy Analytics, Data Science, Machine Learning, Regression, Time-Series Prediction, Energy Management

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CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Electricity use varies with the time of day, the weather, household needs and industrial activity. Historical usage records help us understand consumption patterns and estimate future demand. Typical applications are energy monitoring, consumption planning and efficient energy management.

1.2 PROBLEM STATEMENT

Smart meters log readings every 30 minutes, but raw logs alone do not reveal patterns. Weather, past usage and calendar effects may all matter, yet which of them actually drive consumption is unclear. A data-driven approach is needed to predict consumption and to flag abnormal readings.

1.3 OBJECTIVES

1. To analyse electricity consumption using descriptive statistics.
2. To preprocess the smart-meter dataset.
3. To investigate correlations between consumption, weather and past usage.
4. To build regression and time-series models to predict consumption.
5. To visualise daily, weekly and monthly patterns and detect anomalies.

1.4 SCOPE OF THE PROJECT

DATASET	Kaggle Smart Meter Electricity Consumption Dataset (ziya07).
COVERAGE	5,000 half-hourly readings, 1 January to 14 April 2024; the location is not specified.
PARAMETERS	Consumption, temperature, humidity, wind speed and average past consumption.
APPROACH	Software-only analysis; IoT metering is left as a future extension.

CHAPTER 2

DATASET DESCRIPTION AND REPRESENTATION

2.1 DATA COLLECTION

The data come from the Kaggle Smart Meter Electricity Consumption Dataset (ziya07). It contains **5,000 observations** and **7 columns**, recorded every **30 minutes** from 1 January to 14 April 2024.

2.2 DATASET FEATURES

Table 2.1: Smart-meter dataset features. All numeric columns are min-max scaled, so values are relative, not kWh.

FEATURE	TYPE	DESCRIPTION
Timestamp	Datetime	Reading time, every 30 minutes
Electricity_Consumed	Numerical (0-1)	Target: relative electricity consumption
Temperature	Numerical (0-1)	Scaled ambient temperature
Humidity	Numerical (0-1)	Scaled relative humidity
Wind_Speed	Numerical (0-1)	Scaled wind speed
Avg_Past_Consumption	Numerical (0-1)	Average past consumption level
Anomaly_Label	Categorical	Normal / Abnormal (250 abnormal readings)

2.3 TABULAR REPRESENTATION

Rows correspond to meter readings and columns to measured features. The first five rows of the dataset are shown below.

Table 2.2: First five rows of the dataset.

TIMESTAMP	ELECTRICITY	TEMPERATURE	HUMIDITY	WIND SPEED	AVG PAST	LABEL
2024-01-01 00:00	0.458	0.470	0.396	0.445	0.692	Normal
2024-01-01 00:30	0.352	0.466	0.451	0.459	0.540	Normal
2024-01-01 01:00	0.483	0.285	0.408	0.470	0.615	Normal
2024-01-01 01:30	0.629	0.482	0.512	0.576	0.757	Normal
2024-01-01 02:00	0.336	0.625	0.672	0.373	0.674	Normal

2.4 MATRIX REPRESENTATION

The numeric part of the dataset is represented by a matrix, where m is the number of samples and n the number of features. Here **m = 5,000** samples (rows) and **n = 5** numeric features (columns): consumption, temperature, humidity, wind speed and average past consumption.

$$X = [\, x_{ij} \,] \quad (i = 1..m,\ j = 1..n)$$

CHAPTER 3

DATA PREPROCESSING AND STATISTICAL ANALYSIS

3.1 DATA CLEANING

The dataset is already clean. The checks and the defensive cleaning applied in code are summarised below.

- **Missing values:** 0. All 7 columns are complete.
- **Duplicates:** 0. There are no duplicate rows or timestamps.
- **Timing:** all 4,999 time steps are exactly 30 minutes apart.
- **Abnormal readings:** 250 readings are labelled Abnormal. They are kept, not removed, because they are the anomaly-detection target.

> **Defensive cleaning applied in code:** sort by timestamp, drop duplicate timestamps, interpolate any gaps, then add the Hour, DayOfWeek, Month and IsWeekend features.

3.2 DATA TRANSFORMATION AND NORMALIZATION

All five numeric columns already lie in [0, 1] (minimum 0, maximum 1), so values show relative consumption, not kWh. No further scaling was applied. For reference, min-max normalization is:

$$x' = (x - x_{min}) / (x_{max} - x_{min})$$

Derived features added for modelling:

- **Calendar:** Hour, DayOfWeek, Month, IsWeekend.
- **History:** Lag_1, Lag_2, Lag_48, Roll_Mean_6. Lag_48 is consumption 24 hours earlier (48 half-hour steps).
- **Anomaly:** Anomaly (0/1), Roll_Std_6, Diff_From_Avg.

3.3 DESCRIPTIVE STATISTICS

Table 3.1: Descriptive statistics of the numeric variables.

VARIABLE	MEAN	MEDIAN	STD. DEV.	VARIANCE	MIN	MAX
Electricity_Consumed	0.377	0.377	0.164	0.0269	0.000	1.000
Temperature	0.525	0.524	0.136	0.0184	0.000	1.000
Humidity	0.498	0.497	0.147	0.0215	0.000	1.000
Wind_Speed	0.465	0.465	0.120	0.0145	0.000	1.000
Avg_Past_Consumption	0.459	0.458	0.147	0.0215	0.000	1.000

The mode is omitted because, for continuous scaled values, it carries no useful information. Consumption has mean 0.377 and median 0.377, so its distribution is centred and balanced. Mean consumption is almost equal for Normal (0.377) and Abnormal (0.379) readings, but the standard deviation doubles: 0.152 for Normal versus 0.319 for Abnormal.

3.4 CORRELATION ANALYSIS

The Pearson correlation between two variables X and Y is:

$$r_{XY} = Cov(X, Y) / (\sigma_X * \sigma_Y)$$

Table 3.2: Pearson correlation matrix (5,000 readings).

	CONSUMPTION	TEMPERATURE	HUMIDITY	WIND SPEED	AVG PAST
Consumption	1.00	0.00	0.00	0.00	0.32
Temperature	0.00	1.00	-0.01	-0.01	0.00
Humidity	0.00	-0.01	1.00	0.01	0.01
Wind speed	0.00	-0.01	0.01	1.00	0.00
Avg past	0.32	0.00	0.01	0.00	1.00

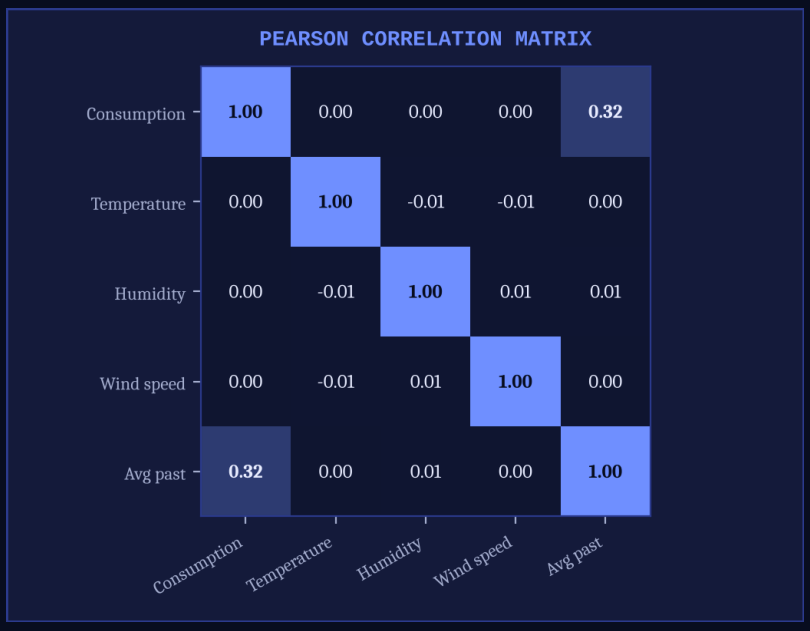


Figure 3.1: Pearson correlation heatmap.

Avg_Past_Consumption is the only meaningful relationship (r = 0.32). With |r| < 0.01, temperature, humidity and wind speed are essentially uncorrelated with consumption, so the weather columns carry little predictive signal in this dataset.

CHAPTER 4

PROPOSED METHODOLOGY

4.1 SYSTEM WORKFLOW

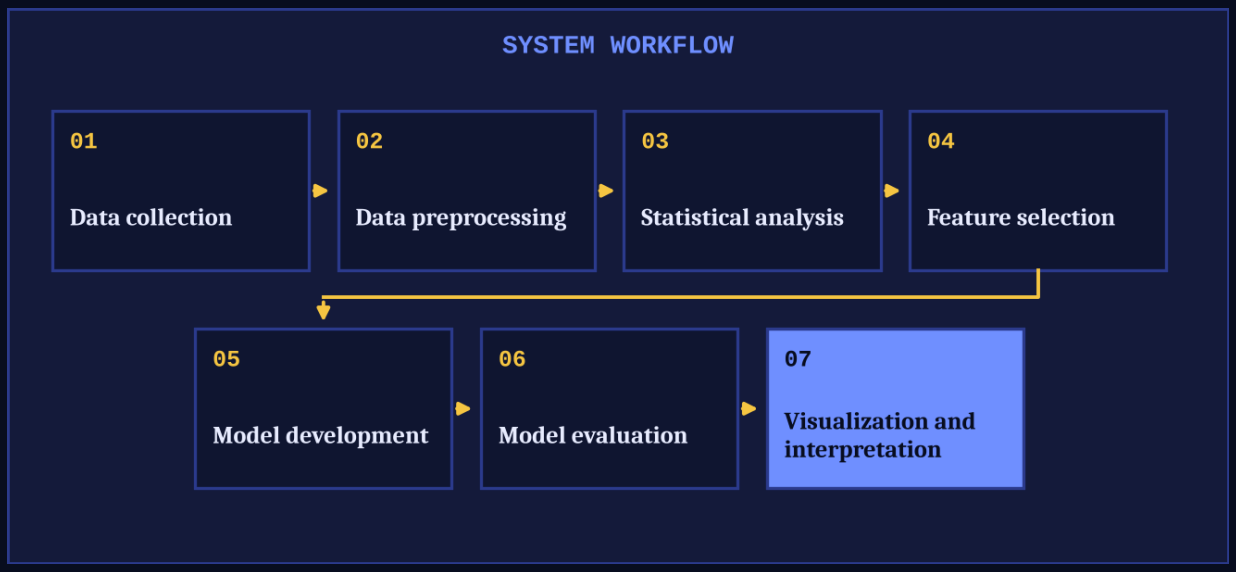


Figure 4.1: System workflow from data collection to interpretation.

1. **Data collection:** smart-meter CSV with 5,000 half-hourly rows.
2. **Data preprocessing:** sort, de-duplicate, interpolate and add time features.
3. **Statistical analysis:** summary statistics and correlations.
4. **Feature selection:** weather, history, lag and calendar inputs.
5. **Model development:** regression, forecasting and anomaly detection.
6. **Model evaluation:** MAE, RMSE and R^2; precision, recall and F1.
7. **Visualization and interpretation:** patterns, forecasts and anomalies.

> **Evaluation principle:** every model is compared with a simple baseline, and the train/test split is chronological, so models are always tested on readings that come after the data they learned from.

4.2 MACHINE LEARNING ALGORITHMS

Table 4.1: Algorithms used for each task.

TASK	ALGORITHM	IDEA
Regression	Linear / Ridge	Weighted sum of features; Ridge penalises large weights
Regression	Random Forest	Average of 200 decision trees
Regression	Gradient Boosting	Small trees that correct earlier errors
Regression	Baseline	Predict the training mean
Time series	Naive	Repeat the value from the same time yesterday
Time series	ARIMA (2,0,1)	Models autocorrelation and short-term shocks
Time series	Decomposition	Checks for a daily cycle (period = 48)
Time series	Baseline	Predict the training mean
Anomaly detection	Isolation Forest	Unsupervised; anomalies are isolated in fewer random splits
Anomaly detection	Random Forest classifier	Supervised; learns from labels with balanced class weights

Anomaly detection is judged by precision, recall and F1-score, not by accuracy alone.

4.3 FEATURE ENGINEERING

- **Weather:** Temperature, Humidity, Wind_Speed.
- **History:** Avg_Past_Consumption, Lag_1 (30 min), Lag_2 (1 h), Lag_48 (24 h), Roll_Mean_6.
- **Calendar:** Hour, DayOfWeek, IsWeekend.

Roll_Mean_6 is the mean of the previous six readings, shifted by one step so that it never uses the current value.

4.4 MODEL TRAINING AND TESTING

The split is **chronological 80 / 20**, so no future information leaks into training. The **training set** has 3,961 rows (2 January to 24 March 2024) and the **test set** has 991 rows (24 March to 14 April 2024).

CHAPTER 5

IMPLEMENTATION

5.1 SOFTWARE AND TOOLS

LANGUAGE	Python 3 in Jupyter Notebook.
LIBRARIES	NumPy, Pandas, Matplotlib, Seaborn.
MACHINE LEARNING	Scikit-learn.
TIME SERIES	Statsmodels.
AI ASSISTANCE	ChatGPT / Gemini / Copilot (optional).
HARDWARE	Laptop, 4 GB RAM or more; no sensors needed.

5.2 IMPLEMENTATION PROCEDURE

1. Load the CSV and inspect shape, types and ranges.
2. Clean: sort, de-duplicate and interpolate.
3. Engineer calendar, lag and rolling features.
4. Compute statistics, correlations and plots.
5. Train and compare models on the 80% split.
6. Evaluate on the 20% test set and interpret the results.

To run the analysis, place `smart_meter_data.csv` in the same folder as the notebook `electricity_consumption_analysis.ipynb`, install the dependencies, and run all cells:

```
pip install numpy pandas matplotlib seaborn scikit-learn statsmodels
```

5.3 SAMPLE CODE

```
df = pd.read_csv("smart_meter_data.csv",
                 parse_dates=["Timestamp"])
df = df.sort_values("Timestamp")

print(df.isnull().sum())   # missing values
print(df.describe())       # statistics
print(df.select_dtypes("number").corr())

for lag in [1, 2, 48]:     # 30 min, 1 h, 24 h
    df[f"Lag_{lag}"] = df["Electricity_Consumed"].shift(lag)

split = int(len(df) * 0.8) # chronological split
```

- `sort_values` keeps readings in time order, which the lags and the split rely on.
- `corr()` builds the correlation matrix used in Chapter 3.
- `shift(lag)` copies earlier consumption into the current row as a predictor.
- The `0.8` split makes the test set strictly later than the training set.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 STATISTICAL RESULTS

Consumption is nearly flat. Hourly means range from 0.357 to 0.398, weekday means from 0.366 to 0.385, and monthly means from 0.370 to 0.383. There is no strong daily or weekly seasonality, unlike real household data, which is consistent with a synthetic dataset. The descriptive statistics and correlations are in Tables 3.1 and 3.2.

6.2 DATA VISUALIZATION

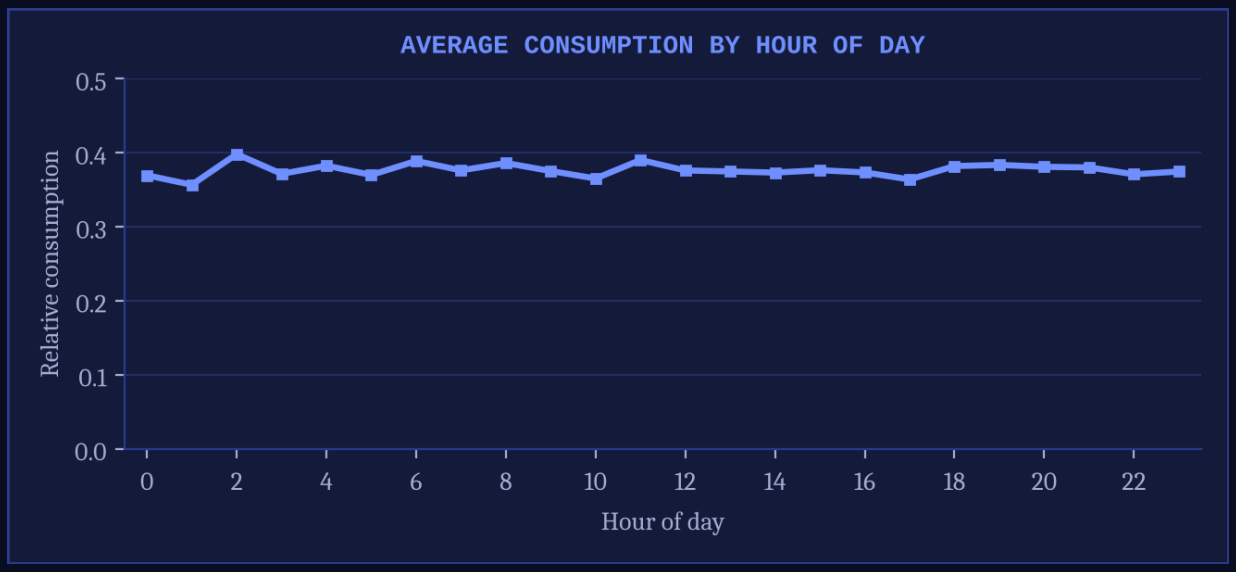


Figure 6.1: Average consumption by hour of day.

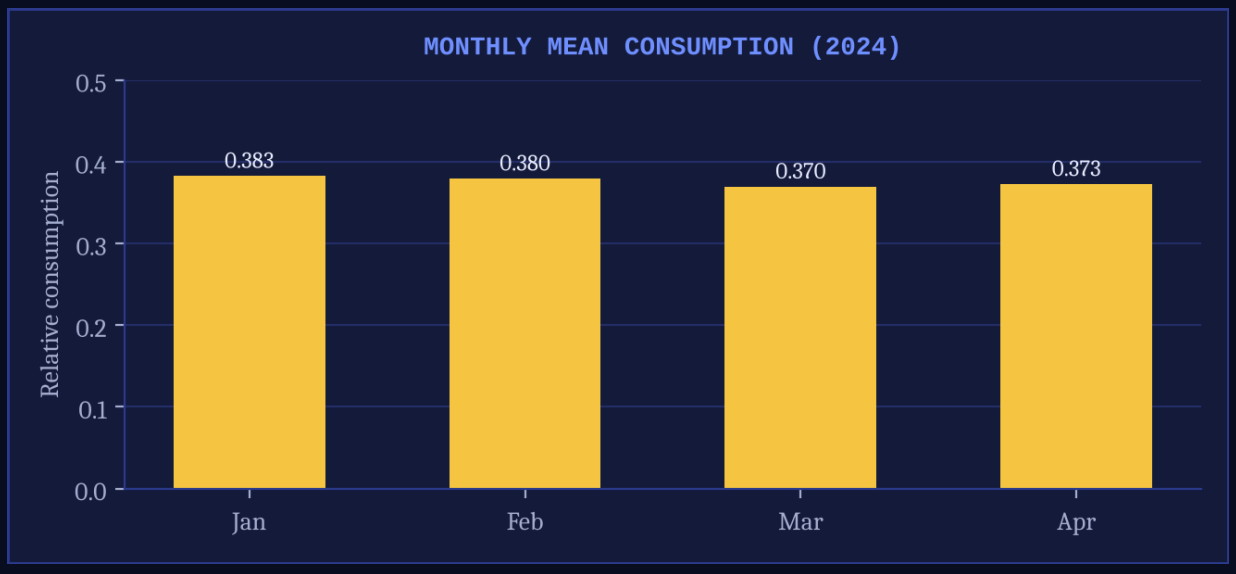


Figure 6.2: Monthly mean consumption (2024).

Both charts start at zero on purpose, so the flat profile is shown at its true scale.

6.3 MODEL EVALUATION

Regression models are scored with MAE, RMSE and R^2. Anomaly detectors are scored with accuracy, precision, recall, F1-score and a confusion matrix:

```
Accuracy  = (TP + TN) / (TP + TN + FP + FN)
Precision = TP / (TP + FP)
Recall    = TP / (TP + FN)
F1-score  = 2 * Precision * Recall / (Precision + Recall)
```

Regression. All models are tested on the chronological test set of 991 readings.

Table 6.1: Regression results on the test set.

MODEL	MAE	RMSE	R^2
Mean baseline	0.130	0.162	-0.004
Linear Regression	0.113	0.142	0.236
Ridge	0.114	0.142	0.232
Random Forest	0.117	0.145	0.195
Gradient Boosting	0.116	0.143	0.217

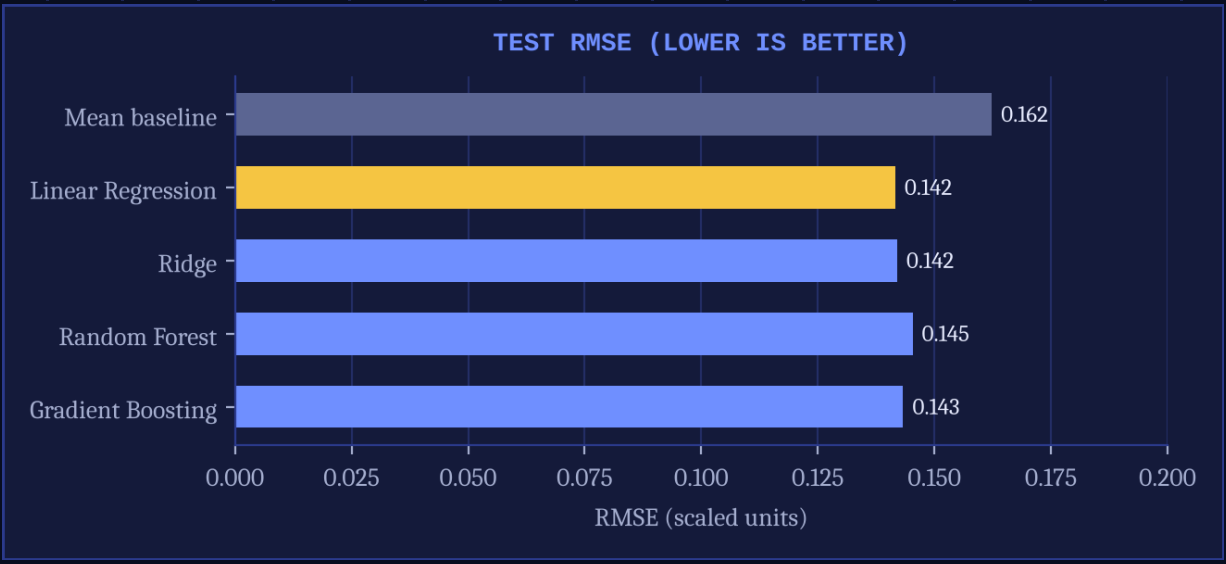


Figure 6.3: Test RMSE by model (lower is better).

Linear Regression is best: R^2 = 0.24 and RMSE 13% lower than the mean baseline. More complex models do not help because the signal is weak.

Feature importance. About 52% of the Random Forest importance comes from Avg_Past_Consumption (0.31) and Roll_Mean_6 (0.22). Temperature, humidity and wind speed contribute about 7% each, a small share spread over noisy inputs. IsWeekend contributes about 0.3%, so weekends do not change consumption here. Weather importance does not imply signal, because the correlations were near zero.

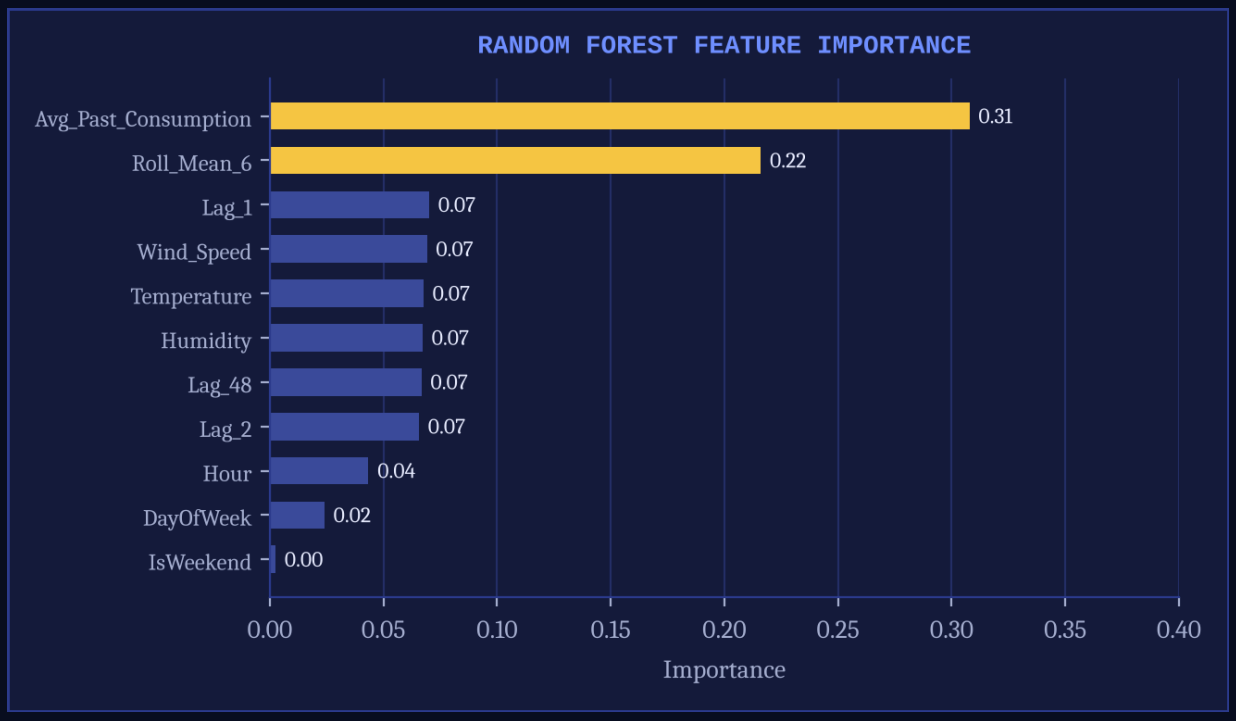


Figure 6.4: Random Forest feature importance.

Time-series forecasting. The last 7 days (336 readings) were forecast with two baselines.

Table 6.2: Forecast error on the last 7 days (336 readings).

BASELINE	MAE	RMSE
Mean baseline	0.130	0.163
Naive (yesterday)	0.188	0.233

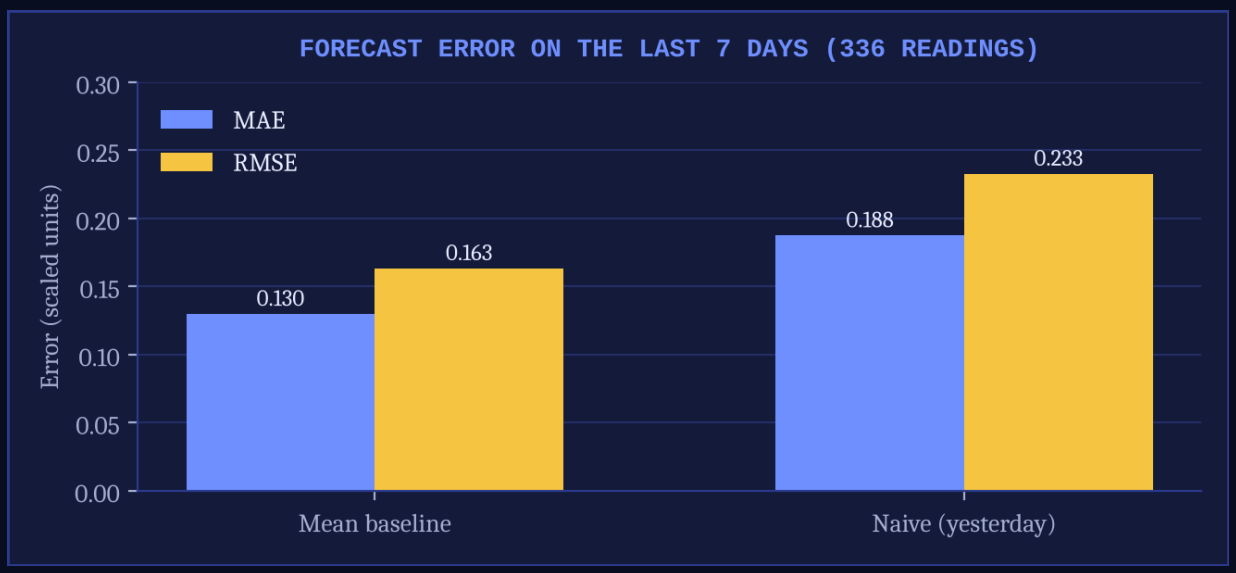


Figure 6.5: Forecast error of the two baselines.

Yesterday is a poor forecast: RMSE rises from 0.163 (mean) to 0.233 when we repeat the same time yesterday. Consecutive days are almost independent, matching the flat hourly profile. ARIMA(2,0,1) and seasonal decomposition are implemented in the notebook (they require statsmodels) but their results are not reported here.

Anomaly detection. The chronological test set has 999 readings (944 Normal, 55 Abnormal).

Table 6.3: Anomaly-detection metrics for the Abnormal class.

MODEL	PRECISION	RECALL	F1-SCORE
Isolation Forest (unsupervised)	0.44	0.36	0.40
Random Forest classifier (supervised)	0.88	0.96	0.92

Table 6.4: Confusion matrices (counts of test readings).

MODEL	ACTUAL CLASS	PRED. NORMAL	PRED. ABNORMAL
Isolation Forest	Normal	919	25
Isolation Forest	Abnormal	35	20
Random Forest classifier	Normal	937	7
Random Forest classifier	Abnormal	2	53

> **Accuracy is misleading:** always predicting Normal scores 95%. Abnormal readings are extreme values (std. dev. 0.32 vs 0.15), which a supervised model can exploit; the labelling rule is undocumented.

6.4 DISCUSSION

Effectiveness

- The project gives a complete, reproducible pipeline from the raw CSV to evaluated models.
- Regression gains consistently over every baseline, though modestly (R^2 = 0.24, RMSE 13% lower).
- Labelled anomalies are detected reliably by a supervised classifier (F1 = 0.92).

Limitations

- About three-quarters of the variance remains unexplained.
- Values are scaled and appear synthetic, so findings may not transfer to real kWh data.
- Only 3.5 months of data are available, so no annual seasonality can be learned.

CHAPTER 7

AI TOOLS AND ETHICAL CONSIDERATIONS

7.1 USE OF AI TOOLS

- Understanding the dataset and its columns.
- Code assistance and debugging.
- Generating analytical summaries.
- Preparing visualizations.
- Report preparation.

7.2 ETHICAL CONSIDERATIONS

DATA RELIABILITY	Scaled, apparently synthetic data should not be generalised.
PRIVACY	Real smart-meter data reveals household routines; anonymise and aggregate it.
RESPONSIBLE AI	Verify AI-generated code and text, and disclose its use.
AUTOMATION LIMITS	False alarms and missed events need human review.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

8.1 CONCLUSION

- The dataset has 5,000 half-hourly readings with no missing values or duplicates.
- Only Avg_Past_Consumption relates to consumption (r = 0.32); weather is near zero.
- There is no strong hourly, weekly or monthly seasonality.
- Linear Regression is the best model (R^2 = 0.24), a modest gain over the baseline.
- Supervised anomaly detection reaches F1 = 0.92; Isolation Forest reaches only 0.40.

8.2 FUTURE SCOPE

- Real-world kWh data with holidays and occupancy.
- Advanced models such as LSTM networks.
- A Streamlit dashboard for monitoring.
- IoT energy meters (ESP32 + current sensor), with proper electrical isolation.

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